

Survival on the Edge: Mountain goat behavioural response to wildfire in southern British Columbia

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BACKGROUND

Climate change is changing fire regimes and **increasing the frequency and intensity of wildfires** globally¹.

British Columbia (BC) is a fire-dominated and highly mountainous landscape that has recently been experiencing **increase in extreme wildfires**². Fire can have direct effects on wildlife by triggering immediate behavioural responses³. However, there is limited research on the behaviour of wildlife **during** a fire³.

Mountain goats (*Oreamnos americanus*) are the **least** studied ungulate⁴ and occur in areas **at risk of extreme wildfires** and are **vulnerable** to climate-driven rapid environmental change⁵. Understanding the **immediate responses of wildlife to wildfires** is critical for predicting how they might respond to future fire regime changes to ensure individuals **can survive wildfire** events.

Given the knowledge gap and importance, we sought to gain insight on:



How do mountain goats respond in real-time to the progression of wildfire?

Do changing properties of the fire affect mountain goat responses?

COARSE-SCALE RESPONSE

Mountain goats had significant decreased their home-range size but maintained similar diffusion rates during the fire period.

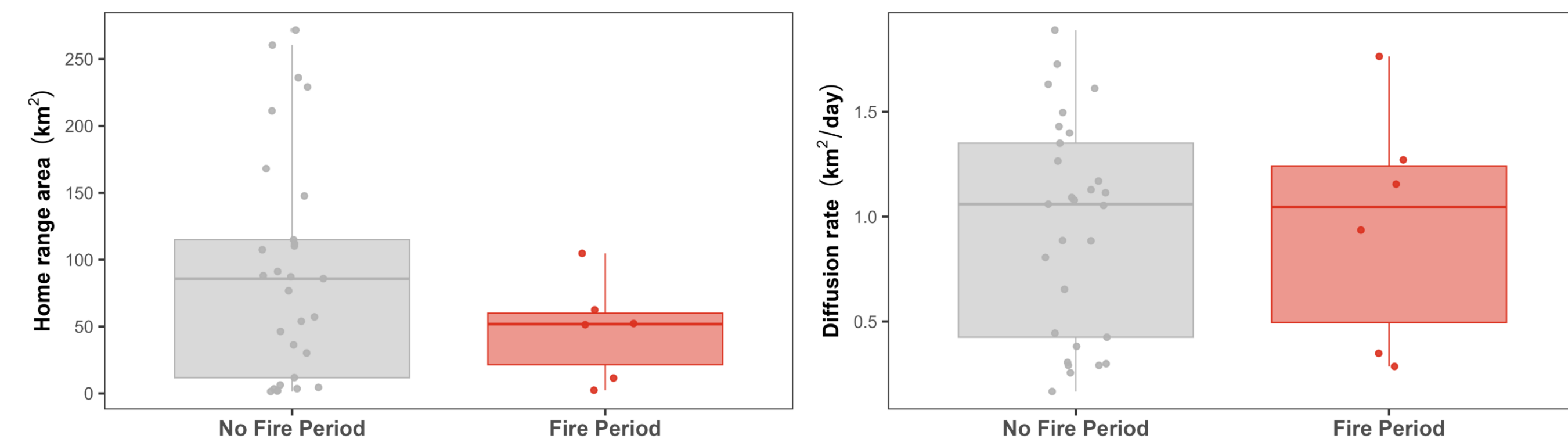


Figure 3. Mountain goat home-range area and diffusion rates. Home-range sizes and diffusion rates during wildfire period (July 22 to October 26) across all years (2019-2024). Red indicates the fire year, and the points depict the underlying individual data.

Mountain goats showed strong habitat selection and redistribution on the landscape during the fire period.

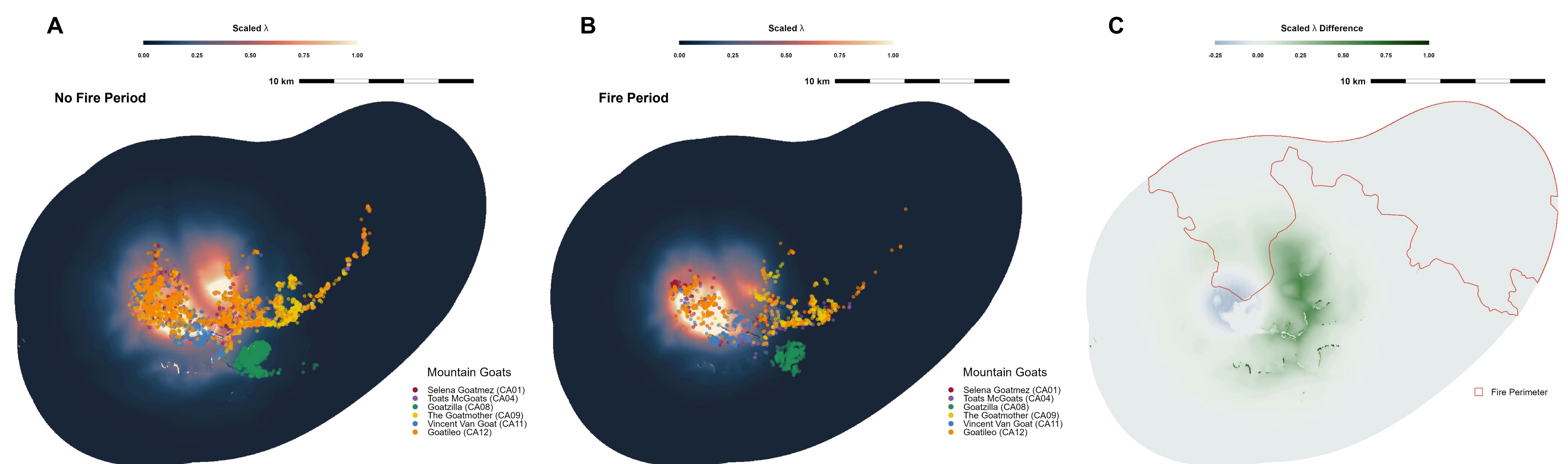


Figure 4. Mountain goat resource selection. Panel (A) shows estimated habitat selection (λ) for the no-fire period. Panel (B) shows estimated habitat selection (λ) for fire period. Panel (C) shows estimated difference habitat selection (λ). Scaled λ values above 0 (green) indicates the higher intensity of use during no fire period, values below 0 (blue) indicates higher intensity of use during the fire period, and 0 (white) denotes no difference between periods.

METHODS

Cathedral Provincial Park

- Within the unceded and ancestral land of the Syilx Okanagan Nation
- Located in the southern interior of BC on the US-Canada border
- In the northeastern region of the Cascade Mountain Range

Crater Creek wildfire (K52125)

- Active fire period: Jul. 22 – Oct. 26, 2023
- Burn duration: 96 days
- Burned size: 46,504 hectares

Mountain goats

- 6 mountain goats
- Continuously GPS-tracked throughout the Crater Creek wildfire
- 6 years of GPS collar data (2019-2024)

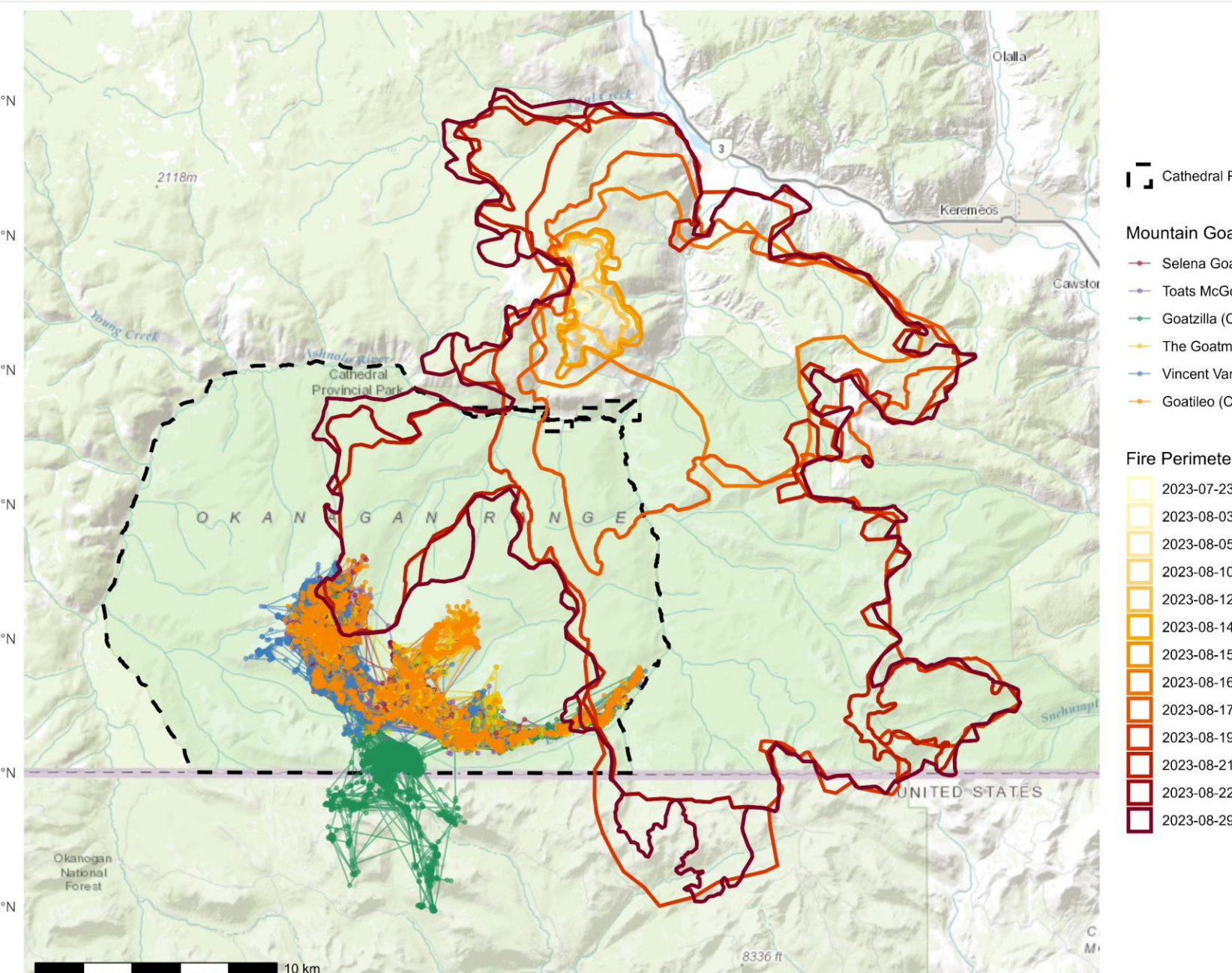


Figure 1. Mountain goat location in and around Cathedral Provincial Park, British Columbia. Map showing GPS tracking data for six mountain goats. The black dashed line indicates the boundary of Cathedral Provincial Park. The light purple line with a dashed line marks the Canada-USA international border. Individual goats are identified by name with collar tag (goat IDs) provided in the parentheses.

We compared mountain goat behaviour between two periods

- No Fire period: Jul. 22 – Oct. 26, 2019-2022 & 2024
- Fire period: Jul. 22 – Oct. 26, 2023

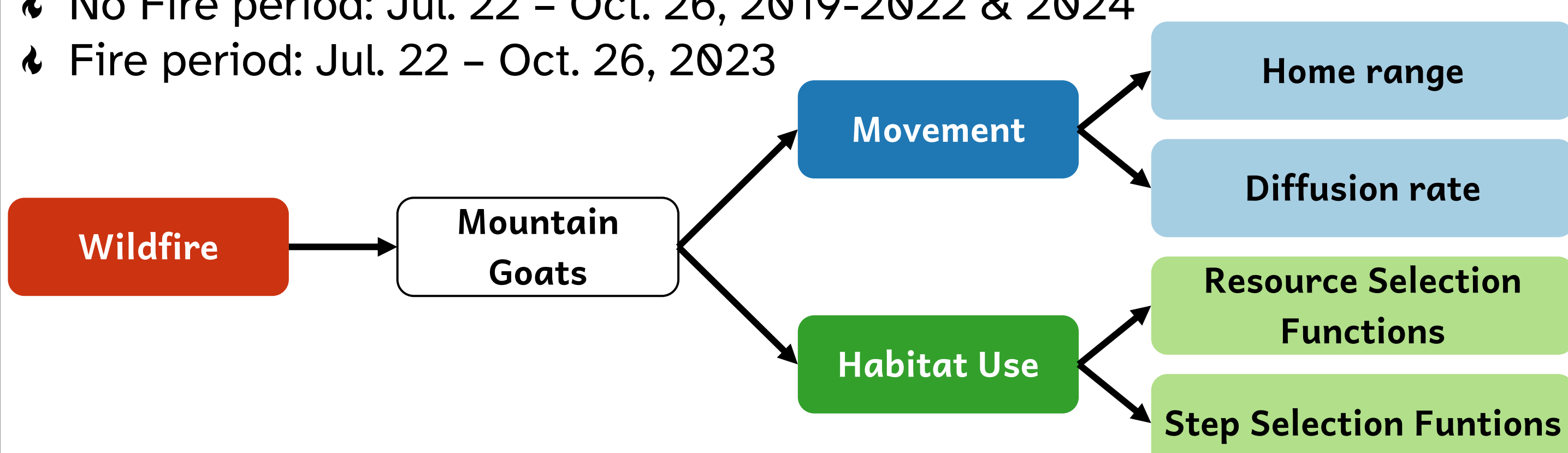


Figure 2. Project schematics of the workflow examining the behavioural responses of mountain goats to wildfire. Movement (blue) was evaluated by home-range size and diffusion rate. Habitat use (green) was investigated based on elevation and distance to steep slopes ($> 50^\circ$).

FINE-SCALE RESPONSE

Habitat selection varied with proximity to the fire with selection for lower elevation when the fire was close and higher elevation when the fire was near.

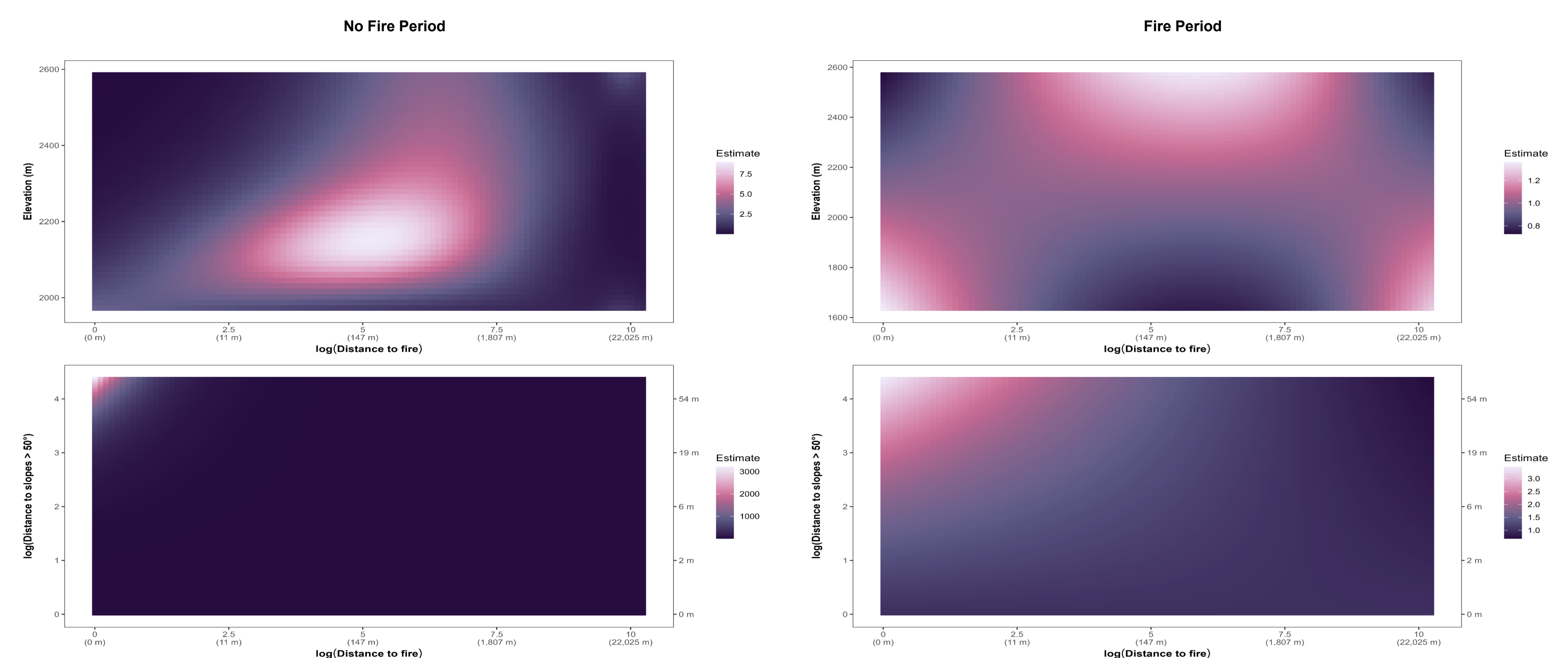


Figure 5. Mountain goat habitat use. Tensor products showing interaction between elevation and distance to fire (top row) and distance to steep slopes ($> 50^\circ$) and distance to fire (bottom row). The left panels represent the no-fire period for 2022, and the right panels represent the fire period. Colours represent the relative selection associated with each of the combination of values. Lighter the colour, the stronger the relative selection and darker the colour, the weaker the relative selection.

TAKE HOME MESSAGES

- All** six mountain goats **survived** the wildfire and remained within their established range.
- Mountain goats were significantly **affected by wildfire**.
- Mountain goats were constricted to a **smaller area**, maintained similar amount of movement but **redistributed** where they were, and **altered their use** in that space based on their proximity to the fire.
- Mountain goat behavioural response reflects **spatial adjustments** during a fire rather than large scale displacement from their home range, emphasising the **importance of connected landscapes**.